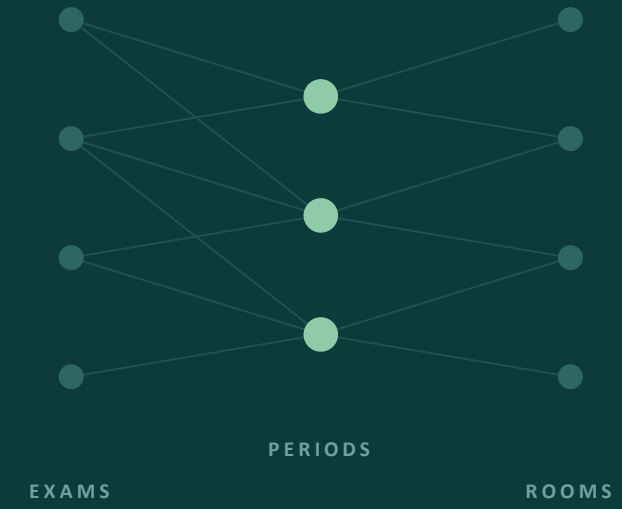
23MNG336 Operations Research – First Review

EXAMOPT



University Examination Timetabling and Room Allocation using Optimization

PROJECT TEAM

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Scheduling Thousands of Exams Without Conflict

- Every semester, universities need to schedule many examinations using a limited number of examination periods and rooms.
- The timetable must satisfy important rules such as avoiding student conflicts, respecting fixed exam assignments, and ensuring that rooms have enough capacity.
- Creating such a timetable manually is difficult because many exams, students, periods, rooms, and constraints have to be considered at the same time.

OUR PROJECT HAS TWO MAIN DECISIONS

? **WHEN should each exam be held**

Assign each exam to an available examination period without creating student conflicts.

? **WHERE should each exam be held**

Assign each scheduled exam to a suitable room while considering capacity, penalties and room splitting.

CORE CHALLENGES

Student Conflicts

No student may have two examinations in the same period.

Fixed Assignments

Certain exams are pre-fixed to specific periods or rooms.

Room Capacity

Every room assignment must respect seating capacity limits.

Distribution Rules

Institutional constraints govern spacing and grouping of exams.

Why Optimization Matters in Examination Scheduling

A good examination timetable reduces student conflicts and uses university resources efficiently, making it an important Operations Research problem.

St

Student Experience

Avoids direct examination conflicts by ensuring that a student does not have two exams in the same period.

Rm

Resource Efficiency

Uses examination rooms efficiently by reducing unused capacity while satisfying room-capacity requirements.

Sc

Scalability

Provides a systematic way to handle large examination scheduling problems with thousands of exams and students.

Bm

Benchmark Relevance

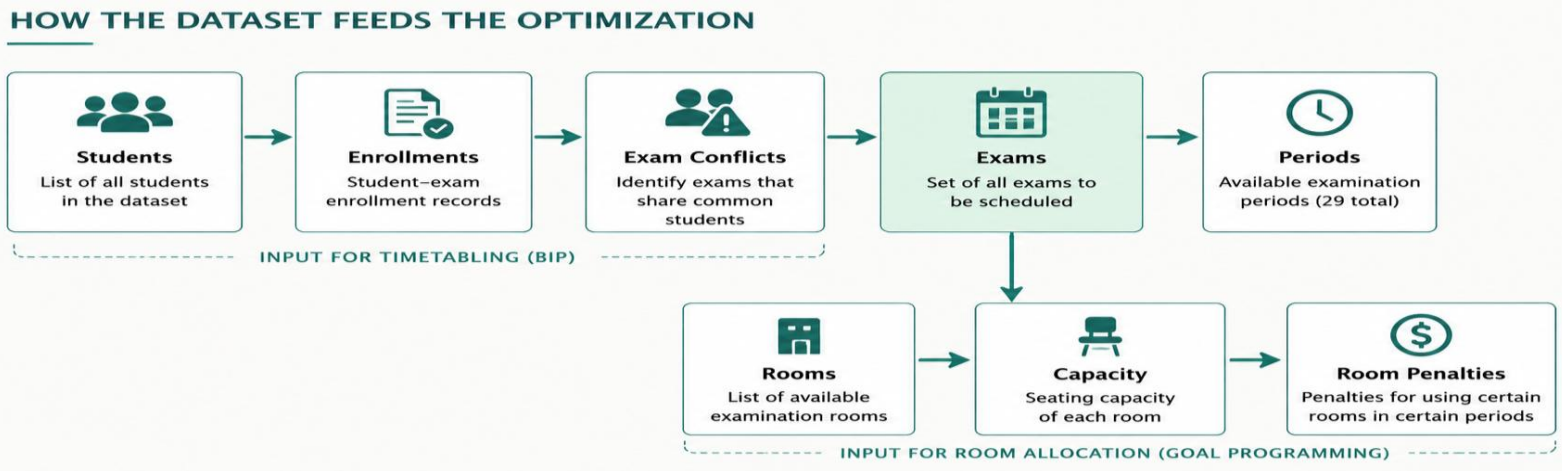
Purdue Fall 2008 provides a realistic large-scale benchmark for testing our optimization models.

Purdue University — Fall 2008 Examination Benchmark

Instance: pu-exam-fal08 · Established examination-timetabling benchmark

2,198 EXAMS	**34,988** STUDENTS	**130,539** ENROLLMENTS
29 EXAMINATION PERIODS	**166** FIXED-TIME EXAMS	**74** FIXED-ROOM EXAMS
13 DISTRIBUTION CONSTRAINTS	**24** LARGE EXAMS	**2** EXAMS REQUIRING ROOM SPLIT

HOW THE DATASET FEEDS THE OPTIMIZATION



Research Questions

RQ1

Examination Timetabling

Can Binary Integer Programming create a feasible timetable by assigning each exam to a suitable period?

RQ2

Student Conflicts

Can the timetable avoid conflicts between exams that have common students?

RQ3

Period Quality

Can the model reduce assignments to undesirable examination periods?

RQ4

Room Allocation

Can Goal Programming assign suitable rooms to all scheduled exams while satisfying room capacity?

RQ5

Resource Utilization

Can we use available room capacity efficiently and reduce unused space?

RQ6

Multi-objective Trade-offs

How do the different room-allocation goals affect the final room assignments?

RQ7

Overall Performance

How well do the BIP and Goal Programming models perform on the Fall 2008 benchmark?

A Two-Stage Optimization Approach

ExamOpt solves the examination scheduling problem in two stages. First, we decide when each exam should be held. Then, we decide which room should be used for each exam.

STAGE 1 · "WHEN?"

Binary Integer Programming

Examination Timetabling

Binary Integer Programming (BIP) is an optimization method where decision variables take only 0 or 1 values. In our project, it is used to decide when each exam should be conducted, while satisfying exam conflicts, fixed-time exams, and distribution constraints.

STAGE 2 · "WHERE?"

Goal Programming

Examination Room Allocation

Goal Programming (GP) is an optimization method used when there are multiple goals that need to be considered simultaneously. In our project, it is used to decide which room should be assigned to each exam, while balancing unused capacity, room penalties, and room splitting.

Proposed Operations Research Methodology



01 Data Validation & Preprocessing

Clean and prepare the Purdue Fall 2008 dataset containing exams, students, enrollments, periods, rooms and constraints.

02 Exam Conflict Matrix

Use student enrollment data to identify pairs of exams that have common students.

03 BIP: Examination Timetabling

Formulate and solve a Binary Integer Programming model to assign each examination to a suitable period.

04 Weighted Goal Programming

Using the BIP timetable, formulate and solve a Weighted Goal Programming model for room allocation.

05 Performance Evaluation

Evaluate the final timetable and room allocation using feasibility, penalty, utilization and computational measures.

The two optimization stages are executed sequentially: the BIP period assignment becomes a fixed input to the Goal Programming room allocation.

Examination Timetabling — Binary Integer Programming

Sets & Parameters

- E — Set of 2,198 exams
- T — Set of 29 examination periods
- C — Conflicting exam pairs (student enrolments)
- FT — Set of 166 fixed-time exams
- D — Set of distribution constraints

Decision Variable

$$x_{e,t} \in \{0, 1\}$$

1 if exam e is assigned to period t

0 otherwise

Objective Function

$$\min \sum_{e \in E} \sum_{t \in T} p_{e,t} \cdot x_{e,t}$$

$p_{e,t}$ = penalty for assigning exam e to period t. If undefined, treat as a feasibility BIP.

Constraints

- $\sum_{t \in T} x_{e,t} = 1 \quad \forall e \in E$ — each exam assigned exactly one period
- $x_{e,t} + x_{e',t} \leq 1 \quad \forall (e,e') \in C, \forall t$ — conflicting exams excluded from the same period
- $x_{e,t} = 1 \quad \forall e \in FT$ — fixed-time exams remain in their prescribed periods
- Distribution constraints $\forall d \in D$ — required spacing / grouping rules are satisfied

Examination Room Allocation — Goal Programming

Sets & Parameters

- E — Set of exams
- R — Set of examination rooms
- t^*_e — Period assigned to exam e by the BIP model
- N_e — Number of students registered for exam e
- C_r — Capacity of room r
- P_r — Penalty for using room r

Decision Variables

$$y_{e,r} \in \{0, 1\} \quad q_{e,r} \geq 0$$

$y_{e,r} = 1$ *if exam e uses room r;* $q_{e,r}$ = *students of e seated in r*

Hard Constraints

- ▶ $\Sigma_r\, q_{e,r} = N_e \quad \forall e$ — all students of an exam are accommodated
- ▶ $q_{e,r} \leq C_r \cdot y_{e,r} \quad \forall e,r$ — room capacity is never exceeded
- ▶ $\Sigma(e: t^*_e=t)\, y_{e,r} \leq 1 \quad \forall r,t$ — a room hosts at most one exam per period

Goals

G1 — **Minimize unused capacity**

$U = \Sigma_{e,r}\,(C_r y_{e,r} - q_{e,r}),\ U + d_1^- - d_1^+ = G1$

G2 — **Minimize room penalties**

$R = \Sigma_{e,r}\, P_r y_{e,r},\ R + d_2^- - d_2^+ = G2$

G3 — **Minimize room splitting**

$S = \Sigma_e\,(\Sigma_r\, y_{e,r} - 1),\ S + d_3^- - d_3^+ = G3$

$$\min Z = w_1 d_1^+ + w_2 d_2^+ + w_3 d_3^+$$

Performance Evaluation Metrics

Planned evaluation metrics for the solved schedule. No numerical results are reported at this stage — these are the metrics that will be measured once the models are solved.

Feasibility

↓

Direct conflicts

Lower is better

↓

Hard constraint violations

Lower is better

↓

Room capacity violations

Lower is better

Timetabling Quality

↓

Period penalty

Lower is better

↓

Objective value

Lower is better

Room Quality

↑

Room utilization

Higher is better

↓

Unused capacity

Lower is better

↓

Room penalty

Lower is better

↓

Room splits

Lower is better

Computational

↓

Solver runtime

Lower is better

9 · EXPECTED OUTCOMES

1

Feasible Examination Timetable

A conflict-free assignment of all 2,198 exams to the 29 available examination periods.

2

Optimized Room Allocation

A room assignment for every scheduled exam, balanced across the three Goal Programming objectives.

3

Comparative Performance Evaluation

Measured feasibility, timetabling-quality, room-quality and computational metrics for the benchmark instance.

4

Interactive ExamOpt Dashboard

A decision-support view summarising the generated timetable, room plan and evaluation metrics.

5

Insight into Trade-offs

A clearer understanding of the trade-offs between unused capacity, room penalties and room splitting.

Planned Tools & Technologies

Py

Python

Core implementation language for data processing and model orchestration.

Op

PuLP / OR-Tools

Modelling layer for the BIP and Goal Programming formulations.

Sv

CBC / Solver Backend

Solves the binary integer and goal-programming optimization models.

Pd

Pandas & NumPy

Cleans and structures the Purdue dataset and builds the conflict matrix.

Vz

Matplotlib / Plotly

Visualises timetable structure, room utilisation and evaluation metrics.

Db

Streamlit / Dash

Powers the interactive ExamOpt decision-support dashboard.

Project Plan & Next Steps

PHASE 1

Data Preparation

Validate the Purdue dataset and construct the student conflict matrix.

PHASE 2

BIP Development

Formulate and solve the timetabling model for Exam → Period.

PHASE 3

Goal Programming Development

Formulate and solve the room allocation model for Exam → Room.

PHASE 4

Evaluation & Dashboard

Compute performance metrics and build the ExamOpt dashboard.

PHASE 5

Reporting & Final Review

Consolidate findings into the final report and presentation.

Thank You